

# Assignment: Teen Depression Prediction using Machine Learning

## Business Problem

A mental health organization wants to identify teenagers who may be at risk of depression based on their lifestyle, social media usage, sleep patterns, stress levels, and behavioral indicators.

Your task is to build a Machine Learning model that predicts whether a teenager is likely to be depressed.

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## Phase 1: Dataset Understanding

### Task 1

Load the dataset and answer the following:

1. Total number of rows and columns.
2. List all features.
3. Identify:
  - Numerical columns
  - Categorical columns
  - Target column
4. Explain the business meaning of each feature.

### Expected Output

Candidate should provide:

- Dataset summary
  - Data dictionary
  - Feature classification
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# Phase 2: Data Cleaning

## Task 2

Perform data preprocessing:

1. Check missing values.
2. Check duplicate records.
3. Handle inconsistent values.
4. Fix data types.
5. Create a clean dataset.

## Questions

1. How many null values exist?
  2. How did you handle them?
  3. Why did you choose that method?
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# Phase 3: Exploratory Data Analysis

## Task 3

Perform EDA and answer:

## Univariate Analysis

Create plots for:

- Age
- Sleep Hours
- Stress Level
- Anxiety Level
- Social Media Usage

## Questions

1. Which feature has the highest variation?
  2. Which feature appears normally distributed?
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## Bivariate Analysis

Create:

1. Sleep Hours vs Depression
2. Social Media Usage vs Depression
3. Stress Level vs Depression
4. Anxiety Level vs Depression

## Questions

1. Which feature has the strongest relationship with depression?
  2. Which feature has the weakest relationship?
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## Correlation Analysis

Create:

- Correlation Heatmap

## Questions

1. Top 5 positively correlated features.
  2. Top 5 negatively correlated features.
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# Phase 4: Feature Engineering

## Task 5

Create new features:

1. High Screen Time

None

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daily_social_media_hours > 5
```

2. Low Sleep

None

`sleep_hours < 6`

3. High Stress

None

`stress_level > 7`

4. High Anxiety

None

`anxiety_level > 7`

## Questions

1. Why is feature engineering important?
  2. Which engineered feature is most useful?
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# Phase 5: Class Imbalance Handling

Your dataset contains:

- Not Depressed  $\approx 97\%$
- Depressed  $\approx 3\%$

## Task 6

Apply:

1. Random Oversampling
2. SMOTE

## Questions

1. Why is accuracy not sufficient here?
  2. What is the difference between SMOTE and Oversampling?
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# Phase 6: Machine Learning Model Building

## Task 7

Build the following models:

1. Logistic Regression
  2. Decision Tree
  3. Random Forest
  4. XGBoost
  5. LightGBM (Bonus)
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## Data Preparation

1. Train-Test Split
  2. Encoding
  3. Feature Scaling
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## Evaluation Metrics

Calculate:

- Accuracy
  - Precision
  - Recall
  - F1 Score
  - ROC-AUC
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## Questions

1. Which model performed best?
2. Which metric is most important and why?
3. Why is Recall important in this project?

Expected Answer:

Because missing a depressed teenager is more dangerous than generating a false alarm.

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# Phase 7: Hyperparameter Tuning

## Task 8

Apply:

None

GridSearchCV

or

None

RandomizedSearchCV

for:

- Random Forest
- XGBoost

## Questions

1. What parameters were tuned?
  2. Did performance improve?
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# Phase 8: Feature Importance Analysis

## Task 9

Identify Top 10 Important Features.

Create:

- Feature Importance Plot

## Questions

1. Which feature contributes most to depression prediction?
  2. Are the findings consistent with EDA?
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## Phase 9: Business Insights

Candidate must provide:

### Key Findings

Example:

- Low sleep increases depression risk.
- High social media usage increases risk.
- Stress and anxiety are strong indicators.

### Recommendations

At least 5 recommendations for parents, schools, or mental health organizations.

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